



INT221

MVC PROGRAMMING

Lecture #0

The kick start session



Course Details

- Course Code:- INT221
- LTP:- 2 0 2 [2 lectures, 2 practical's /week]
- Credits:- 3



Vision

To be a globally recognized school through excellence in teaching, learning and research for creating Computer Science professionals, leaders and entrepreneurs of future contributing to society and industry for sustainable growth.



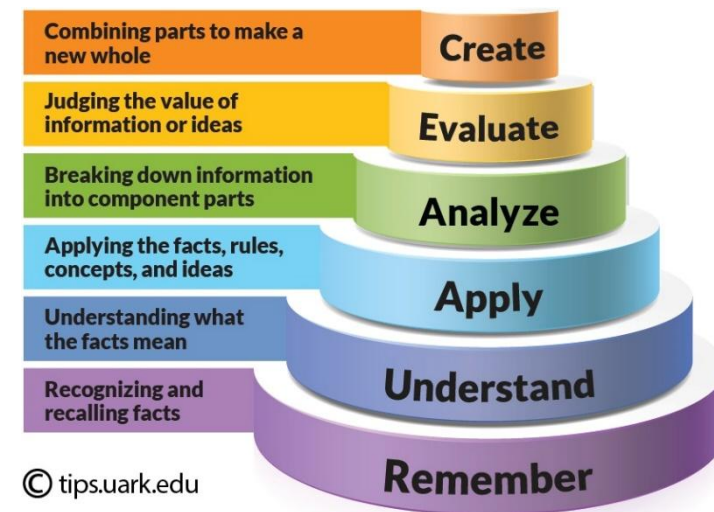
Mission

- To build computational skills through hands-on and practice-based learning with measurable outcomes.
- To establish a strong connect with industry for in-demand technology driven curriculum.
- To build the infrastructure for meaningful research around societal problems.
- To nurture future leaders through research-infused education and lifelong learning.
- To create smart and ethical professionals and entrepreneurs who are recognized globally

Revised Bloom's Taxonomy

Bloom's Taxonomy is a framework for categorizing educational goals, It is widely used by educators to create learning objectives, assessments, and activities that encourage higher-order thinking. The taxonomy is typically represented as a hierarchical model with six levels:

- **Remembering:** Recalling facts and basic concepts.
- Example: Define, list, memorize, repeat, state.
- **Understanding:** Explaining ideas or concepts.
- Example: Describe, explain, interpret, summarize, classify.
- **Applying:** Using information in new situations.
- Example: Use, execute, implement, demonstrate, solve.
- **Analyzing:** Breaking information into parts to explore understandings and relationships.
- Example: Compare, contrast, examine, test, differentiate.
- **Evaluating:** Justifying a decision or course of action.
- Example: Critique, judge, argue, support, validate.
- **Creating:** Producing new or original work.
- Example: Design, construct, plan, produce, invent.



Course outcome

Through this course students should be able to:

CO1 :: Recall features of Laravel and the installation process.

CO2 :: Apply routing, handle HTTP requests, and customize responses for seamless Laravel application navigation.

CO3 :: Develop controllers, utilize Blade templates, and apply advanced routing for modular and structured development.



Course outcome

Through this course students should be able to:

CO4 :: Manage cookies, emails, sessions for convenient and robust functionality.

CO5 :: Design forms and execute form validation to ensure data integrity.

CO6 :: Employ databases to streamline data handling and enhance functionality in Laravel.

Program Outcomes

PO1

Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2

Problem analysis::Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3

Design/development of solutions::Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4

Conduct investigations of complex problems::Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Program Outcomes

PO5

Modern tool usage::Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6

The engineer and society::Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7

Environment and sustainability::Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8

Ethics::Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9

Individual and team work::Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Program Outcomes

PO10

Communication::Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11

Project management and finance::Demonstrate knowledge and understanding of the engineering, management principles and apply the same to one's own work, as a member or a leader in a team, manage projects efficiently in respective disciplines and multidisciplinary environments after consideration of economic and financial factors.

PO12

Life-long learning::Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Course contents

Unit I

Getting started with MVC Laravel framework : What is MVC framework?, Overview of Laravel framework and its features, Introduction to composer, Latest Composer installation, Latest Laravel installation, Directory/Application structure, Artisan.

Unit II

Request, Routing & Responses : Laravel Request Lifecycle, Basic Routing, Routing Parameters, Understanding Views in Laravel, Passing Data to Views, Sharing Data with all Views, Laravel Response- Attaching Headers, Attaching Cookies, JSON Response, Laravel Redirections, Redirecting to Named Routes, Redirecting to Controller Actions.

Course contents

Unit III

Controllers, Blade and Advanced Routing : Basic Controllers- Creating Controllers, Controller Routing, Controller Middleware, Restful Resource Controllers, Blade- Creating Templates, PHP Output, Controller Structures, Templates Inheritance, Advanced Routing- Named Routes, Secure Routes, Parameter Constraints, Route Groups, Route Prefixing, Domain Routing, URL Generation- The current URL, Generating Framework URLs, Asset URLs, Generation Shortcuts

Course contents

Unit IV

URL Generation, Request Data and Emails : Request Data-Retrieval, Old Input, Uploaded Files, Cookies, Sending Emails.

Laravel Localization and Sessions : Laravel Localization and Examples, Laravel Session- Accessing Session Data, Storing Session Data, Deleting Session Data.

Unit V

Laravel Form validation : CSRF field, Method field, Laravel Form validation, Validation Rules, Error Messages, Custom Validation Rules, Repopulating Forms.

Course contents

Unit VI

Getting started with databases : Model creation, Migrations, CRUD using Query Builder, Seeding, Using MongoDB with Laravel, CRUD using Eloquent ORM(Object Relational Mapper), Implementing Rest APIs.

List of practical's

- Installation of Laravel and Composer using command prompt
- Implement sharing of data using views
- Use route parameters and parameter constraints
- Implement redirections using routes and controllers
- Create and register middlewares in Laravel
- Create blade templates, controller structures and template inheritance in Laravel.
- Use restful resource controllers to perform CRUD operations
- Working with advanced routing and URL methods in Laravel

List of practical's

- Send emails and upload files in Laravel
- Access, store and delete cookie and session data in Laravel.
- Create form and implement simple validation, error messages, custom validation rules and custom validation messages in Laravel.
- Connect to database in Laravel
- Create, modify, update tables in Laravel
- Manage database with migrations
- Create, modify, update tables in Laravel
- Manage database with migrations



Assessment/Evaluation Scheme

- Attendance: 5%
- CA: 45%
- MTT: No MTT
- ETP: 50%

Assessment/Evaluation Scheme

- CA1: 30 marks (Visual Implementation)
- CA2: 30 marks (Project)
- Note: No MOOC Certification is Applicable

Rubrics for CA1 (Visual Implementation)

- Written (15 Marks)
- Viva (5 Marks)
- Code Semantics and Implementation(10 Marks)



CA 2 – Project Rubrics

- Functionality (5),
- User Interface(5),
- Code Quality, Implementation and Presentation-10
- Video Presentation-5 Marks
- Social media presence- 5 Marks(Likes(2 Marks(100+ Marks)),Shares(1 Mark (50+ Shares)),Comments(2 Marks) (30+ Comment))

CA 2 – Project Rubrics For Revenue Generation

- Functionality & User Interface (5),
- Code Quality, Implementation and Presentation-10
- Revenue Generation -10 Marks ($\geq 5K$)
- Social media presence- 5 Marks(Likes(2 Marks(100+ Marks)),Shares(1 Mark (50+ Shares)),Comments(2 Marks) (30+ Comment))

Books

Reference Book:

- **ULTIMATE LARAVEL FOR MODERN WEB DEVELOPMENT**
by **DRISHTI JAIN, AVA PUBLISHING**



Software's

Standard Editor-

1. VS-Code.
2. Composer
3. XAMPP Server



EDU Rev Benefits in RPL (Eligibility to Apply)

Course Code	Sources of Prior Learning	Web URL of Source
INT221	Student having A+ Grade in INT220, Student Needs to Appear and clear ETP of INT 221 with minimum B+ grade to apply for RPL	
INT221	PHP WITH LARAVEL FOR BEGINNERS - BECOME A MASTER IN LARAVEL	https://www.udemy.com/course/php-with-laravel-for-beginners-become-a-master-in-laravel/

Note : Udemy Courses are paid courses, Students can pay to enroll for these courses

EDU Rev Benefits in RPL (After Application)

- Eligibility for Full Course Waiver- (Minimum B+ Grade in the the Exam(ETT) Same Pattern)
- The Exam date will be communicated after application of RPL and Approval for RPL.

EDU Rev Benefits in Project

Course Code	Title of Project	Source	Course Equivalence	Web URL
INT221	Web Development Projects to solve Social Issues or Live Projects solving Problems related to SDG	Smart Indian Hackathon Final Round Merit Holders	Complete Course	http://www.sih.gov.in

Student's Project must be equivalent to the skill set learnt in this course, Hackathons can be conducted by Universities of National Importance

GRADE UPGRADATION

Course Code	CATEGORY OF MAPPING	DESCRIPTION	EQUIVALENCE
INT221	TECHNICAL COMPETITIONS/HACKATHONS	Web Development Projects (Problem Statements) to solve social issues or Live Projects solving the problem related to SDG, Students need to score Merit Position in those Competition.	GRADE UPGRADATION
INT221	COPYRIGHTS/PATENTS – PROBLEM STATEMENTS SHOULD BE MAPPED WITH SIH	Copyright of the web development projects under continuous Assessment(Project Category : problem statements should mapped with SIH)	GRADE UPGRADATION

Education Revolution

25th Academic Session of
LPU

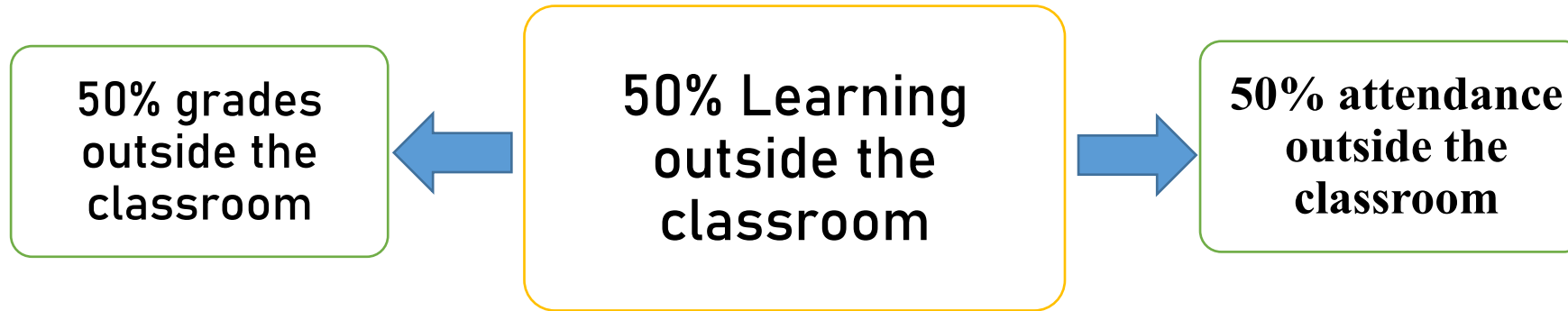
Why EDU–Revolution!!!

- Career Readiness
- Enhance Real World Skills
- Promote Innovation and Creativity
- Foster Industry Relations
- Develop Entrepreneurial Mindset
- Increase Student Engagement and Motivation
- Promote Lifelong Learning

Education Revolution: an approach that encourages...

- Gamified Learning with AI focus - Learning Outside Classroom
- Revenue Generation by Students
- Career (not just degree)
- Students in Decision Making

Pillar I: Gamified Learning with AI focus



Formula: 50-50-50

Techniques of Learning Outside the Classroom

Projects

**Technical Competitions &
Hackathons**

Social Media Presence

**Recognition of Prior
Learning**

Paid Internship

**Recruitment/Comp
etitive/ Eligibility
Examination**

NPTEL/MOOCs

**Community
Service**

**Publications/Patent
s/ Copyrights**

**Projects****Technical Competitions &
Hackathons****Social Media Presence****Recognition of Prior
Learning****Paid Internship****Recruitment/Comp
etitive/ Eligibility
Examination****NPTEL/MOOCs****Community
Service****Publications/Patent
s/ Copyrights**

School to add relevant school specific examples of each category which students have taken in the past or students may take up in the coming term as mentioned above

Gamified Learning: Benefits

Course
Exemption

Grade
Upgradation

No Pen-Paper
based
Evaluation

Attendance
Relaxation

Duty Leaves

Value Addition
in Academic
Transcript

Pillar II: Revenue Generation

Where would Revenue Generation come from?

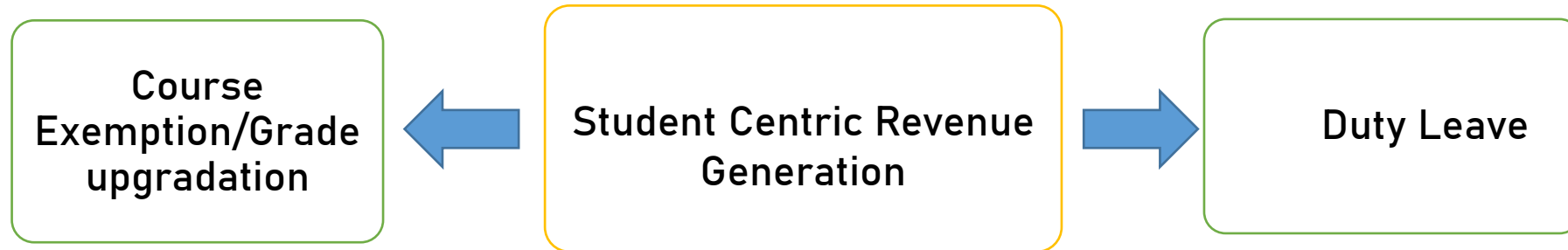
- ✓ Industry/field consultancy projects
- ✓ Freelancing activities
- ✓ Paid internships
- ✓ Establishing start- ups and entrepreneurial venture

Through University support:

- i. Earn while Learn opportunities
- ii. Incentives for Publication/Patent/Copyright

You should be able to generate at least 50 %
age of your semester
fee every semester

Pillar II: Revenue Generation



Important Points:

- Earn exemptions from your revenue generation endeavors
- Earn Academic benefits on revenues above 50,000 (in 6 months)

- Submit authentic information only
- Student must submit a detailed proposal outlining:
 - i. Revenue Model
 - ii. Market Feasibility
 - iii. Execution Strategy
 - iv. Financial Projections

Pillar II: Revenue Generation **School Specific Success Stories**

(Highlighting maximum revenue generated and sustainable revenue generation projects)



Pillar III: Career Pathway

"Your Journey, Your Choice: Build, Serve, or Study for Tomorrow!"

Corporate Jobs

Government
Jobs

Higher Studies

Entrepreneurship

Research

Study Abroad

Pillar III: Career Pathway

- Career Pathways are at the core of our academic framework.
- Curriculum is strategically designed to align with needs and requirements of each career pathway.
- From technical proficiency to soft skills, the curriculum is tailored to develop a well-rounded professional ready for competitive market as per the chosen pathway.

Pillar III: Career Pathway

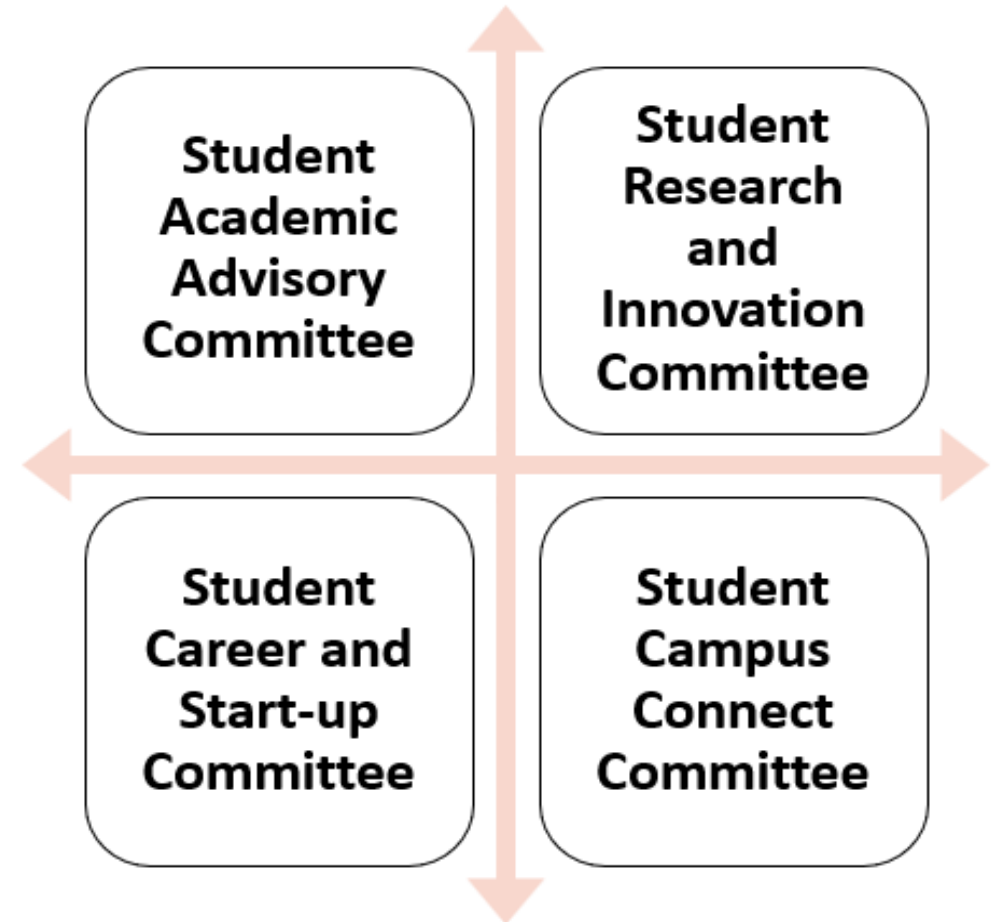
School Specific Success Stories for each pathway

(Highlighting student and their achievement for each pathway)

Pillar IV: Students in Decision Making

Committees under Student Education Revolution Board

We bring numerous ideas every year;
but now, **at least 20% of these ideas**
should originate from students
themselves in all categories like
academics, placements, research,
entrepreneurship, etc.



Pillar IV: Students in Decision Making

A platform to bring the Change

